



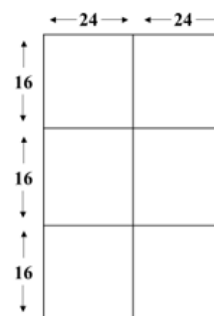
- Suppose in the data of problem 41, we say that HCF of any pair of number is h and the maximum possible number of elements in set is 4, then the maximum possible value of h is **(HCF) NIMCET-2007**
(a)2 (b)3 (c)4 (d)None
- In a group there are 13 boys and 7 girls with brown hair. 15 person don't have brown hair. Then the maximum number of persons the group can have, is **(HCF) NIMCET-2007**
(a)26 (b)25 (c)39 (d)None
- The minimum number of tiles of size 16 by 24 required to form a square by placing them adjacent to one another is **(HCF) NIMCET-2012**
(a) 6 (b)8 (c)11 (d)16
- The integers 34041 and 32506, when divided by a 3- digit integer n , leave the same remainder. What can be the value of n ? **(HCF) NIMCET-2019**
(a)289 (b)307 (c)367 (d)493
- The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively, is **(HCF) NIMCET-2019**
(a)123 (b)127 (c)235 (d)305
- The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5, respectively is **(HCF) NIMCET-2022**
(a)127 (b)123 (c)235 (d)305

ANSWER

1	d	2	d	3	a	4	b	5	b
6	a								

SOLUTION:

- (d)** None
- (d)** 13 boys and 7 girls
15 person don't have brown hair
 $13+7+15= 35$ maximum no. Of persons
- (a)** From option (a),
Let the number of tiles = 6
 $\therefore$ Total length = 48
and total breadth = 48



Since, length = breadth

$\therefore$ Number of tiles form a square = 6

- (b)** Let the common remainder be x . Then numbers $(34041 - x)$ and $(32506 - x)$ would be completely divisible by n .
Hence, the difference of the numbers $(34041 - x)$ and $(32506 - x)$ will also be divisible by n .
 $34041 - x - (32506 - x)$
 $= 34041 - x - 32506 + x$
 $= 1535$
Factors of 1535: 1, 5, 307
Here, 307 is a three digit factor of 1535.
So, $n = 307$
34041 and 32506 divided by 307 leaves remainder 271.
Hence, the value of n is 307.

- (b)** Required number
 $=$ HCF of $(1657 - 6)$ and $(2037 - 5)$
 $=$ HCF of 1651 and 2032 = 127
- (a)** Required number = HCF of $(1657 - 6)$ and $(2037 - 5)$
 $=$ 2032

By division method,

$$\begin{array}{r}
 1651 \overline{)2032} 1 \\
 \underline{1651} \\
 381 \\
 \underline{381} \\
 0 \\
 \hline
 \end{array}$$